



UNIVERSITÀ DI PISA

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Outline

1. Cuprates and Hubbard physics
2. The Hartree-Fock method
3. The HF method and symmetry breaking mechanisms
4. The normal phase
5. The antiferromagnetic phase
6. The superconducting phase
7. Conclusions and outlook

Cuprates and Hubbard physics

Nobel Prize in Physics 1987



Photo from the Nobel Foundation archive.

J. Georg Bednorz

Photo share: 1/2



Photo from the Nobel Foundation archive.

K. Alexander Müller

Photo share: 1/2

Possible High T_c Superconductivity in the Ba–La–Cu–O System

J.G. Bednorz and K.A. Müller
IBM Zürich Research Laboratory, Rüschlikon, Switzerland

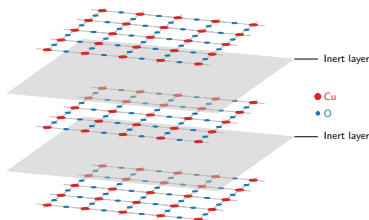
Received April 17, 1986

What makes cuprates interesting?

- ◇ Strong “unconventional” insulators;
- ◇ Rich phase diagram;
- ◇ Origin of Cooper pairing: still unclear;
- ◇ Highest critical temperatures for SC transition in this class of materials.

These materials are stackings of:

- ◇ 2D copper oxide active layers;
- ◇ intermediate layers of inert material.



Most of the cuprates physics is well described by the single-band Hubbard model on the square lattice:

$$\hat{H}_{\text{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \quad \text{with } t, U > 0.$$

The phase diagram of cuprates

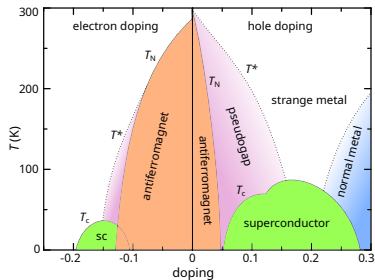
Within cuprates phase diagram:

- ◇ **Antiferromagnetism** at null and low doping;
- ◇ ***d*-wave superconductivity** as we inject or remove charge.

Both arise from strong electronic correlations.



Failure of band theory in describing insulation.



Schematic phase diagram of cuprates. Author: Holger Motzkau, [Wikimedia commons](#), CC BY-SA 3.0.

What is the origin of the insulation mechanism?

Strong $e-e$ interactions and **big energetic cost** of having two electrons on the same site.

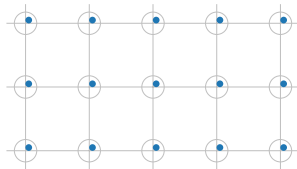
Strongly coupled Hubbard model:

$$U \gg t.$$



Mott insulation mechanism

Electrons are **localised** as in a sort of “traffic jam”.



Mott localisation: a many-body insulator

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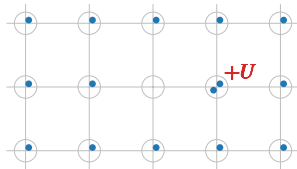
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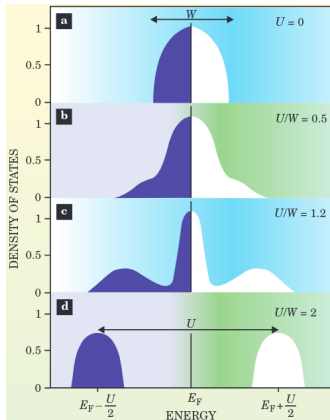


Mott insulation mechanism

Electrons are **localised** as in a sort of “traffic jam”.



Bands flattening: a signature of Mott localisation



[Unsure: move to backup?]

- (a) Free model, conventional DoS;
- (b) Weak coupling, large coherent peak;
- (c) Intermediate coupling, **narrow coherent peak**;
- (d) Strong coupling, lower and upper Hubbard bands.



Localisation = flattening of bare bands.

Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

How does Mott localisation relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Mott localisation at half-filling.



Singlet pairing $\leftrightarrow$ AF ordering.



HM at strong coupling:

$$\hat{H}_{\text{HM}} \simeq \frac{4t^2}{U} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E,$$

Superconductivity

Condensation of Cooper pairs with *d*-wave spatial structure.

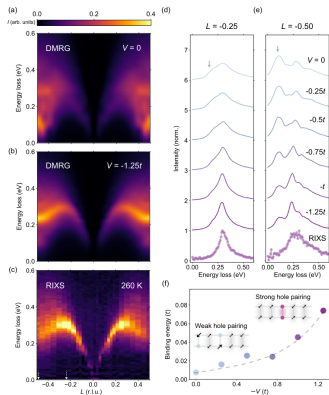


Order parameter $\psi(\mathbf{r}_1 - \mathbf{r}_2)$: antisymmetric upon $\pi/2$ rotations.



Suppressed on-site pairing.

Recent developments: the extended Hubbard model



Hari Padma et al., 2025. *Physical Review X* 15 (2): 021049.

Incompatibilities with the spectrum of magnetic excitation in cuprates.



Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\text{EHM}} = \hat{H}_{\text{HM}} - V \underbrace{\sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V}.$$

$\hat{H}_V$ can act as a **source of d -wave superconductivity**.

The model (EHM):

$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} - \underbrace{V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \quad \text{with } t, U, V > 0.$$

The motivation: Compatibility with recent experimental findings.

The idea: we know that the EHM can host d -wave superconductivity. Can we obtain Mott physics?

The method: theoretical and numerical Hartree-Fock technique.

The Hartree-Fock method

The constrained minimisation framework

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0 .

Variational principle:

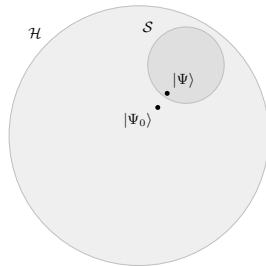
$$\forall |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} : E[\Phi] \geq E_0.$$

$\Downarrow$

Constrained minimisation procedure

$$|\Psi\rangle : \left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi\rangle} = 0.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

Hartree-Fock method: we identify $\mathcal{S}$ with the set of all the ground states of non-interacting hamiltonians,

$$\mathcal{S} = \left\{ |\Psi\rangle \mid |\Psi\rangle \text{ is the ground-state of some } \hat{H}_{\text{HF}} \right\},$$

where

$$\hat{H}_{\text{HF}} \equiv \sum_{\alpha} E_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} \hat{\gamma}_{\alpha} \quad \text{with} \quad \left\{ \hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger} \right\} = \delta_{\alpha\beta},$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

$\Downarrow$

Wick's theorem:

$$\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\delta} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\gamma} \rangle \langle \hat{c}_{\beta}^{\dagger} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\alpha}^{\dagger} \hat{c}_{\beta}^{\dagger} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}}.$$

The self-consistency mapping

Target parametric hamiltonian:

$$\hat{H}_{\text{HF}}[\mathbf{v}] \equiv \sum_{\alpha} E_{\alpha}[\mathbf{v}] \hat{\gamma}_{\alpha}^{\dagger}[\mathbf{v}] \hat{\gamma}_{\alpha}[\mathbf{v}] \quad \text{with ground state } |\Psi[\mathbf{v}]\rangle,$$

being $\mathbf{v}$ a vector of variational parameters (HFPs). HF approximation:

$$|\Psi_0\rangle \simeq |\Psi[\mathbf{v}]\rangle,$$

with $\mathbf{v}$ such that:

Variational picture

Energy is minimised,

$$\left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle=|\Psi[\mathbf{v}]\rangle} = 0.$$

Self-consistency picture

A self-consistency problem is satisfied,

$$A(\mathbf{v}) = \mathbf{v},$$

being A a non-linear map **uniquely determined**.

The iterative algorithm (iHF)

[Unsure: move to backup?] The HF solution is the **fixed point** of a non-linear map A ,

$$A(\mathbf{v}) = \mathbf{v}.$$

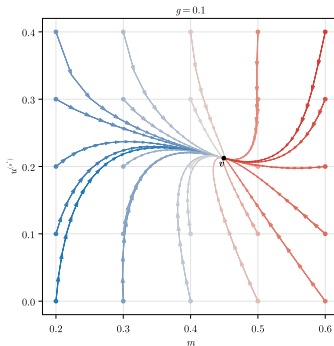
$\Downarrow$

Strategy: “follow” the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv gA(\mathbf{v}_i) + (1 - g)\mathbf{v}_i.$$

with $i > 0$, $g \in [0, 1]$.

Example path (AF) ($t = 1.0$, $U = 3.0$, $V = 6.0$, $L = 128$, $\beta = 100.0$, $\delta = 0.0$)



Example path for the AF phase.

The HF method and symmetry breaking mechanisms

Phase transition: spontaneous breaking of some symmetries of the model.

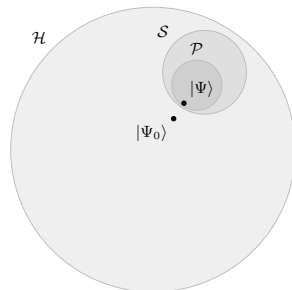
Phase X : shows symmetries $Y, Z \dots$



Further constraining of HF subspace to $\mathcal{P} \subset \mathcal{S}$, states with symmetries $Y, Z \dots$



Symmetry selection rules.



e.g.

$$\text{Charge is conserved} \implies \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle = 0$$

The model (EHM):

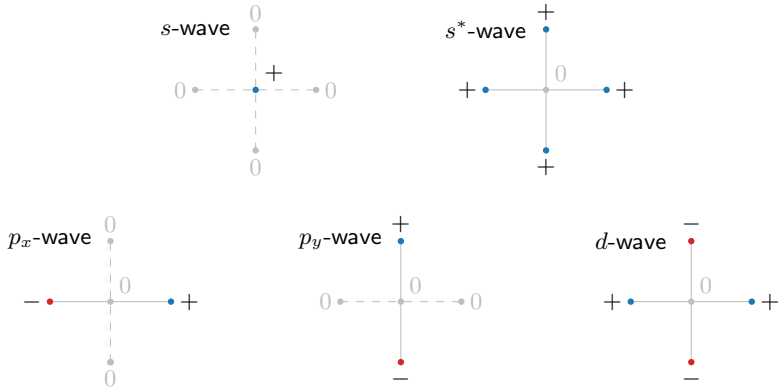
$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} - \underbrace{V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \quad \text{with } t, U, V > 0,$$

and its symmetries:

- ◊ *Gauge invariance* (charge conservation), $U(1)$.
If broken: superconductivity.
- ◊ *Global spin rotational invariance*, $SU(2)$.
If broken: magnetism.
- ◊ *Translational invariance* in directions x and y , $T_1^{(x)} \otimes T_1^{(y)}$.
If broken: non-uniform charge and spin densities.
- ◊ *Rotational invariance* under $\pi/2$ rotations, C_4 .
If broken: nematicity.

The spatial symmetry channels

It is useful to introduce five symmetry structures:



The normal phase

Normal phase: no broken symmetries. From the Fock channel,

$$\tilde{t}_{ij\sigma} \equiv \begin{cases} t - V \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle & \text{if } i, j \text{ are NNs,} \\ 0 & \text{otherwise.} \end{cases}$$

$\Downarrow$

Self-consistency problem: one HFP,

$$u^{(s^*)} = (s^*\text{-wave component of } u_{\mathbf{k}} \equiv \langle \hat{n}_{\mathbf{k}} \rangle),$$

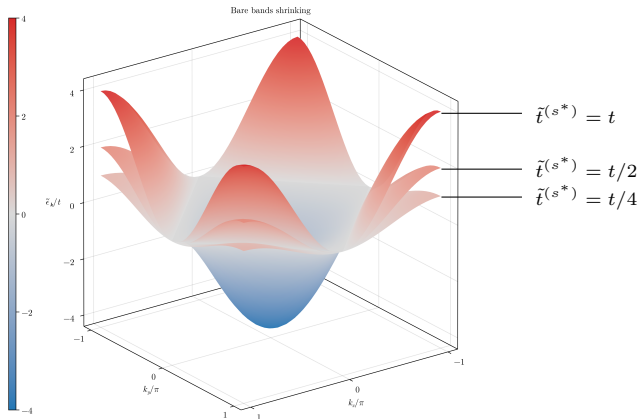
s.t. hopping amplitude is isotropically renormalised,

$$t \rightarrow \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)} V}{2}.$$

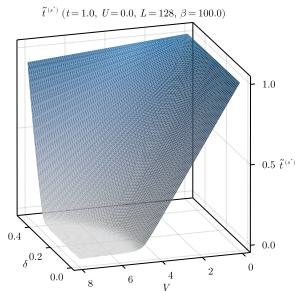
Therefore: renormalisation of electronic bands, $\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma}$.

Theoretical constraint:

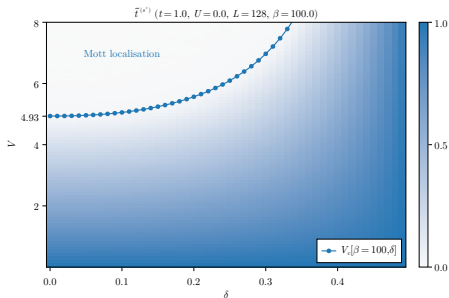
$0 < \tilde{t}^{(s^*)} < t \implies$ **Flattening of bands** = Mott localisation.



In (δ, V) plane, we have determined theoretically the presence of a Mott critical boundary $V_c[\beta, \delta]$,



(a) Numerical results for $\tilde{t}^{(s^*)}$



(b) Critical Mott boundary.

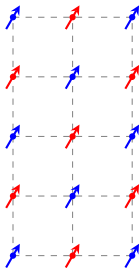
Within the Mott plateau, electrons are completely localised.

The **normal phase**, summarised:

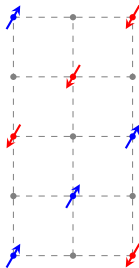
- ◇ Preserves all symmetries of the EHM;
- ◇ The HF approximation is **independent of U** ;
- ◇ Presence of V -induced Mott localisation and **bands flattening**;
- ◇ We identify a Mott transition boundary as we vary the doping level.

The antiferromagnetic phase

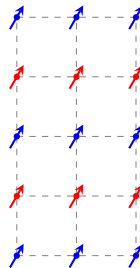
Alternating



Doubled Periodicity



Stripes



Antiferromagnetism in the conventional Hubbard model

The superconducting phase

Conclusions and outlook

That's all, thank you!

Backup slides
